Patient Health Record Consent Management System

Lab 3 – Written Justification

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SECTION- B

Architecture Selection: Chose Layered Architecture for the Patient Health Record Consent Management System.

Architectural Choice: Layered Architecture separates the user-facing portals, application services, and data storage into clear layers. This makes the system easier to understand, maintain, and test.

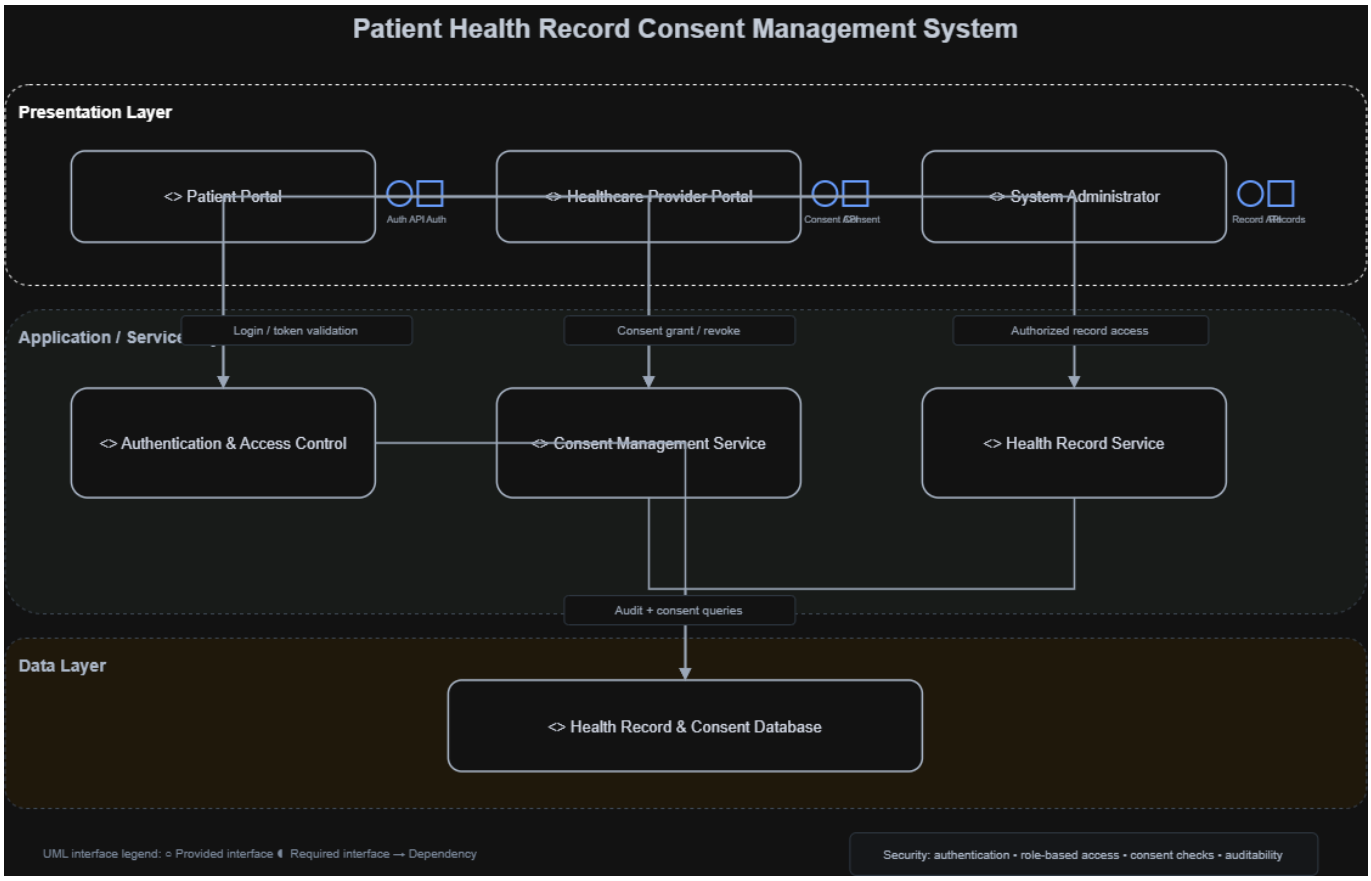
Two Reasons:

1. Healthcare record access and consent processing have distinct responsibilities, so separating presentation, service logic, and data storage reduces coupling and keeps each component focused.

2. Consent rules and health-record operations can be changed in the service layer without redesigning the patient/provider interfaces or the database structure.

Security Advantage: The service layer can enforce authentication, role based access control, explicit consent checks, and auditing before allowing access to protected health records. This provides a central point for consistent security enforcement.

Performance Benefit: Frequently used application services can be optimized independently, while database queries can be indexed and controlled at the data layer. This supports efficient retrieval of authorized records.


Patient Health Record Consent Management System
Presentation Layer
<> Patient Portal
Auth API Auth
<> Healthcare Provider Portal
<> System Administrator
Application / Service
Login / token validation
Consent grant / revoke
Authorized record access
<> Authentication & Access Control
<> Consent Management Service
<> Health Record Service
Audit + consent queries
Data Layer
<> Health Record & Consent Database
UML interface legend: ○ Provided interface ◖ Required interface ⟶ Dependency
Security: authentication • role-based access • consent checks • auditability